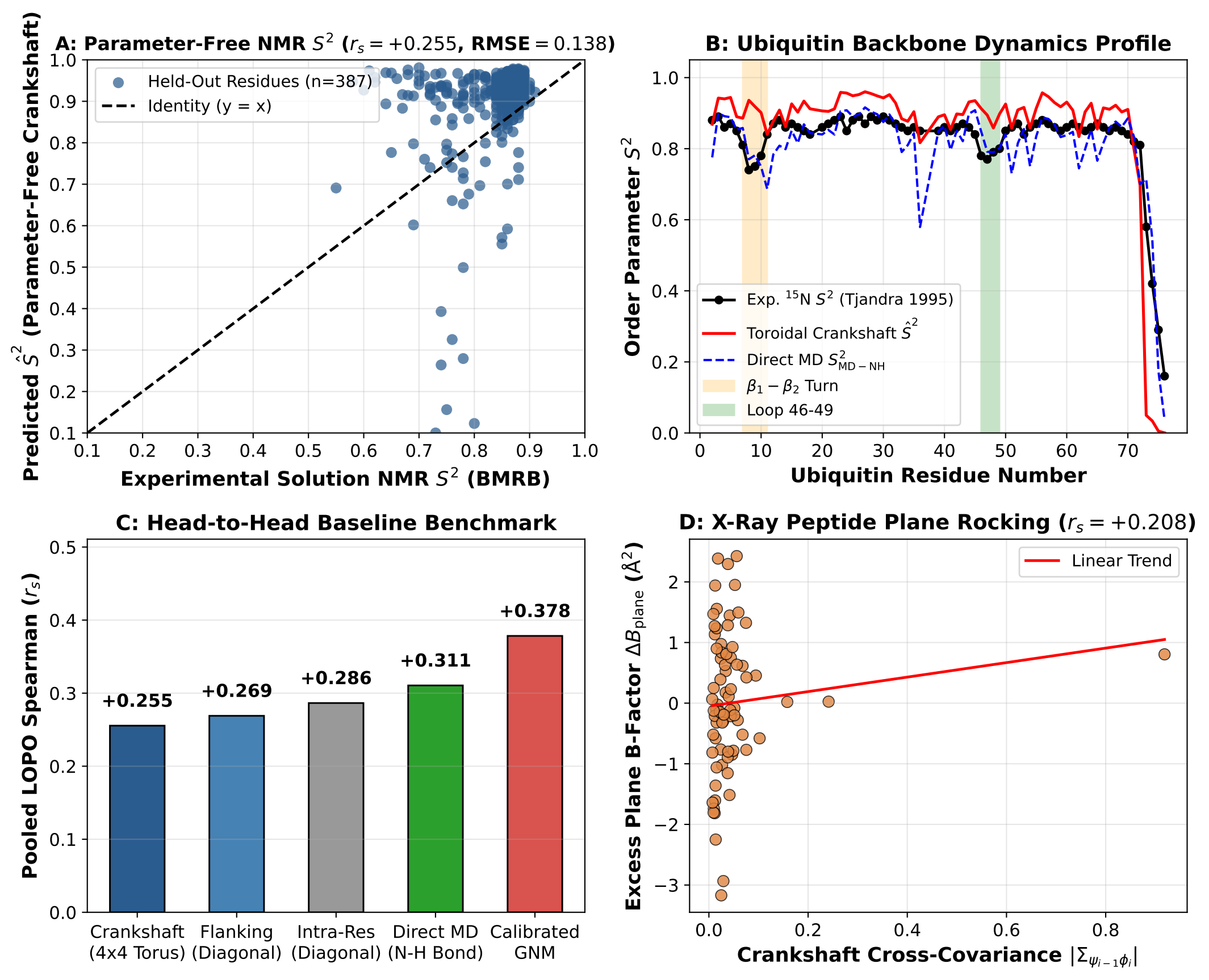

A: Parameter-Free NMR S^2 ($r_s = +0.255$, RMSE = 0.138)
Held-Out Residues (n=387)
Identity (y = x)
Predicted $\hat{S}^2$ (Parameter-Free Crankshaft)
Experimental Solution NMR S^2 (BMRB)
B: Ubiquitin Backbone Dynamics Profile
Exp. ^{15}N S^2 (Tjandra 1995)
Toroidal Crankshaft $\hat{S}^2$
Direct MD S^2_{MD-NH}
$\beta_1 - \beta_2$ Turn
Loop 46-49
Order Parameter S^2
Ubiquitin Residue Number
C: Head-to-Head Baseline Benchmark
Pooled LOPO Spearman (r_s)
+0.255
+0.269
+0.286
+0.311
+0.378
Crankshaft (4x4 Torus)
Flanking (Diagonal)
Intra-Res (Diagonal)
Direct MD (N-H Bond)
Calibrated GNM
D: X-Ray Peptide Plane Rocking ($r_s = +0.208$)
Linear Trend
Excess Plane B-Factor ΔB_{plane} (Å^2)
Crankshaft Cross-Covariance $|\Sigma_{\psi_{i-1}\phi_i}|$